Iterative Model Free Safe Exploration using Event-triggered Impulsive Perturbations - Appendix

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1. Appendix A

A.1 Proof of Lemma 1

1) This condition is proved by its own definition.

2) This condition is proved by having the value function constructed as a sum of positive definite functions $r(\cdot,\cdot)$.

3) Let $J_{C_0}(\boldsymbol{x}_{k+1}) - J_{C_0}(\boldsymbol{x}_k) < 0 \Leftrightarrow J_{C_0}(\boldsymbol{x}_{k+1}) < J_{C_0}(\boldsymbol{x}_k) \Leftrightarrow J_{C_0}(\boldsymbol{x}_{k+1}) < r(\boldsymbol{x}_k, C_0(\boldsymbol{x}_k)) + \gamma J_{C_0}(\boldsymbol{x}_{k+1}) \Leftrightarrow J_{C_0}(\boldsymbol{x}_{k+1}) - \gamma J_{C_0}(\boldsymbol{x}_{k+1}) < r(\boldsymbol{x}_k, C_0(\boldsymbol{x}_k)) \Leftrightarrow (1-\gamma) < \frac{r(\boldsymbol{x}_k, C_0(\boldsymbol{x}_k))}{J_{C_0}(\boldsymbol{x}_{k+1})} \Leftrightarrow$

$$\gamma > \frac{J_{C_0}(\boldsymbol{x}_{k+1}) - r(\boldsymbol{x}_k, C_0(\boldsymbol{x}_k))}{J_{C_0}(\boldsymbol{x}_{k+1})}. \tag{18}$$

This proves the lemma.

A.2 Proof of Lemma 2

We first establish the lower bound $\alpha(\|\boldsymbol{x}_k - \boldsymbol{x}_k^G\|) \le J_{C_0}(\boldsymbol{x}_k)$. With $r(\cdot,\cdot)$ a positive definite function, there exist $\lambda_{min} > 0$ such that $\lambda_{min}\|\boldsymbol{x}_k - \boldsymbol{x}_k^G\| \le r(\boldsymbol{x}_k, C_0(\boldsymbol{x}_k))$. Having $r(\boldsymbol{x}_k, C_0(\boldsymbol{x}_k)) \le J_{C_0}(\boldsymbol{x}_k)$ from (8), we get $\lambda_{min}\|\boldsymbol{x}_k - \boldsymbol{x}_k^G\| \le J_{C_0}(\boldsymbol{x}_k)$, which defines our continuous $\mathcal{K}$-class function $\lambda_{min}\|\boldsymbol{x}_k - \boldsymbol{x}_k^G\| \coloneqq \alpha(\|\boldsymbol{x}_k - \boldsymbol{x}_k^G\|)$.

To prove the upper bound $J_{C_0}(\boldsymbol{x}_k) \le \beta(\|\boldsymbol{x}_k - \boldsymbol{x}_k^G\|)$, we use condition 3) from Lemma 1 together with (8), to get

$$\begin{aligned} J_{C_0}(\boldsymbol{x}_k) &= r(\boldsymbol{x}_k, C_0(\boldsymbol{x}_k)) + \gamma J_{C_0}(\boldsymbol{x}_{k+1}) \\ &\le r(\boldsymbol{x}_k, C_0(\boldsymbol{x}_k)) + \gamma J_{C_0}(\boldsymbol{x}_k) \\ &\Leftrightarrow (1-\gamma) J_{C_0}(\boldsymbol{x}_k) \le r(\boldsymbol{x}_k, C_0(\boldsymbol{x}_k)) \\ &\Leftrightarrow J_{C_0}(\boldsymbol{x}_k) \\ &\le \frac{1}{(1-\gamma)} r(\boldsymbol{x}_k, C_0(\boldsymbol{x}_k)). \end{aligned} \tag{19}$$

There exists $\lambda_{max} > 0$ such that $r(\boldsymbol{x}_k, C_0(\boldsymbol{x}_k)) \le \lambda_{max}\|\boldsymbol{x}_k - \boldsymbol{x}_k^G\|$. Having $J_{C_0}(\boldsymbol{x}_k) \le (1/(1-\gamma)) r(\boldsymbol{x}_k, C_0(\boldsymbol{x}_k))$ from (19), we get $J_{C_0}(\boldsymbol{x}_k) \le (\lambda_{max}/(1-\gamma))\|\boldsymbol{x}_k - \boldsymbol{x}_k^G\|$, which defines our continuous $\mathcal{K}$-class function $(\lambda_{max}/(1-\gamma))\|\boldsymbol{x}_k - \boldsymbol{x}_k^G\| \coloneqq \beta(\|\boldsymbol{x}_k - \boldsymbol{x}_k^G\|)$. This proves the lemma.

A.3 Proof of Theorem 1

Let $k_j^+ = k_j + 1$ be the first time sample after a perturbation impulse that makes $J_{C_0}(\boldsymbol{x}_k) > a(1-b)^k$, and $k_j^- = k_j$ be the first time sample after k_j^+ such that $J_{C_0}(\boldsymbol{x}_k) < a(1-b)^k$. A potential trajectory of system (1) is depicted in Fig. 1 for demonstration purposes. The first impulse time k_1 is shown to happen at time $k_1 = 2$, where $J_{C_0}(\boldsymbol{x}_{k_1}) < a(1-b)^{k_1}$. The impulse time k_1 destabilizes the system, having thus time step k_1^+ such that $J_{C_0}(\boldsymbol{x}_{k_1^+}) > a(1-b)^{k_1^+}$. Namely,

$$k_1^+ = \min\{\, k \in (k_1, \infty) \colon J_{C_0}(\boldsymbol{x}_k) > a(1-b)^k \}. \tag{20}$$

After k_1^+, the system evolves p-steps based on $F(\boldsymbol{x}_k, C_0(\boldsymbol{x}_k))$, until $J_{C_0}(\boldsymbol{x}_k) < a(1-b)^k$ for $k = k_1^-$. Let

$$k_1^- = \min\{k \in (k_1^+, \infty) \colon J_{C_0}(\boldsymbol{x}_k) < a(1-b)^k\}. \tag{21}$$

Then, based on Assumption 2 we have

$$\begin{aligned} J_{C_0}(\boldsymbol{x}_{k_1^-}) &\le c^p J_{C_0}(\boldsymbol{x}_{k_1^+}) \le mc^p J_{C_0}(\boldsymbol{x}_{k_1}) \le mc^p a(1-b)^{k_1} \\ &= mc^p a(1-b)^{k_1} \frac{(1-b)^{k_1^- - k_1}}{(1-b)^{k_1^- - k_1}} \\ &= \frac{mc^p}{(1-b)^{k_1^- - k_1}} a(1-b)^{k_1^-} \end{aligned} \tag{22}$$

Therefore, if $mc^p/(1-b)^{k_1^+ - k_1} < 1$, then the value of $J_{C_0}(\boldsymbol{x}_{k_1^-})$ does not exceed $a(1-b)^{k_1^-}$ at time k_1^-. This is equivalent of saying that the effect of the perturbation impulse from time k_1 is damped at time k_1^- under rule (11). For any general noise event time k_j, we have $k_j^+ = \min\{\, k \in (k_j, \infty) \colon J_{C_0}(\boldsymbol{x}_k) > a(1-b)^k\}$ and $k_j^- = \min\{k \in (k_j^+, \infty) \colon J_{C_0}(\boldsymbol{x}_k) < a(1-b)^k\}$. Using the same proof steps as above, we can conclude that $J_{C_0}\left(\boldsymbol{x}_{k_j^-}\right) \le \frac{mc^p}{(1-b)^{k_j^- - k_j}} a(1-b)^{k_j^-}$, showing that $J_{C_0}\left(\boldsymbol{x}_{k_j^-}\right)$ settles below the threshold $a(1-b)^{k_j^-}$ after the exploratory noise at time k_j. As $a(1-b)^k \to 0$ we have $J_{C_0}(\boldsymbol{x}_k) \le a(1-b)^k \to 0$, proving that the system in closed loop with control rule (11) is both stable and attractive. Then, based on Lemma 2, we get $\|\boldsymbol{x}_k - \boldsymbol{x}_k^G\| \to 0 \Leftrightarrow \boldsymbol{x}_k \to \boldsymbol{x}_k^G$, showing that the system is asymptotically stable to $\boldsymbol{x}_k^G$ in Ω_X under (11), as $k \to \infty$.

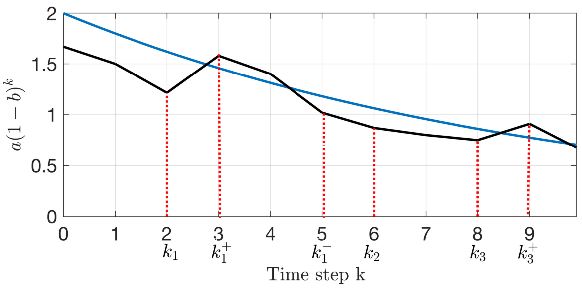

Figure 1. Visual presentation of J_{C_0} function evolution scenario for $T_s = 1$ and $\varrho = 2$: target Lyapunov signal (blue); $J_{C_0}(\boldsymbol{x}_k)$ (dark). The impulse times generated by (11) are $\{k_1, k_2, k_3\}$: k_1 and k_3 are destabilizing the system, while the one from k_2 is stabilizable.

A.4 Proof of Lemma 3

Let $\boldsymbol{x}_{k_j+1}^l = F\left(\boldsymbol{x}_{k_j}, C_0\left(\boldsymbol{x}_{k_j}\right) + \mu_l\left(\boldsymbol{x}_{k_j}, J_{C_0}\left(\boldsymbol{x}_{k_j}\right), V_l\right)\right)$, and $\boldsymbol{x}_{k_j} \in \Psi$. The Lipschitz inequality during the additive impulsive perturbation is defined as

$$\left\|\boldsymbol{x}_{k_j+1}^{l+1} - \boldsymbol{x}_{k_j+1}^l\right\| \le L_F \left\| C_0\left(\boldsymbol{x}_{k_j}\right) + \mu_{l+1}\left(\boldsymbol{x}_{k_j}, J_{C_0}\left(\boldsymbol{x}_{k_j}\right), V_l\right) - C_0\left(\boldsymbol{x}_{k_j}\right) + \mu_l\left(\boldsymbol{x}_{k_j}, J_{C_0}\left(\boldsymbol{x}_{k_j}\right), V_l\right)\right\| = L_F \left\|\mu_{l+1}\left(\boldsymbol{x}_{k_j}, J_{C_0}\left(\boldsymbol{x}_{k_j}\right), V_l\right) - \mu_l\left(\boldsymbol{x}_{k_j}, J_{C_0}\left(\boldsymbol{x}_{k_j}\right), V_l\right)\right\|. \quad (23)$$

For sequential trials l and $l+1$, for a linear variance increase according to $\Theta(\cdot)$ we get

$$\left\|\mu_{l+1}\left(\boldsymbol{x}_{k_j}, J_{C_0}\left(\boldsymbol{x}_{k_j}\right), V_l\right) - \mu_l\left(\boldsymbol{x}_{k_j}, J_{C_0}\left(\boldsymbol{x}_{k_j}\right), V_l\right)\right\| \le \varphi. \quad (24)$$

This results in

$$\left\|\boldsymbol{x}_{k_j+1}^{l+1} - \boldsymbol{x}_{k_j+1}^l\right\| \le L_F \varphi. \quad (25)$$

Given that $\varphi \le d_{min}\left(\boldsymbol{x}_{k_j+1}^l, \Xi\right)/L_F$, we obtain $\left\|\boldsymbol{x}_{k_j+1}^{l+1} - \boldsymbol{x}_{k_j+1}^l\right\| \le d_{min}\left(\boldsymbol{x}_{k_j+1}^l, \Xi\right)$. This proves the lemma.

A.5 Proof of Theorem 2

For any exploration trial l, we define the one-step reachable set for a state $\boldsymbol{x}_{k_j}$ as the set of all possible transitions from $\boldsymbol{x}_{k_j}$ under $C_0\left(\boldsymbol{x}_{k_j}\right) + \mu_l\left(\boldsymbol{x}_{k_j}, J_{C_0}\left(\boldsymbol{x}_{k_j}\right), V_l\right)$, namely

$$X_l^+\left(\boldsymbol{x}_{k_j}\right) = \left\{\boldsymbol{x}_{k_j+1} \,\middle|\, \boldsymbol{x}_{k_j+1} = F\left(\begin{matrix}\boldsymbol{x}_{k_j}, C_0\left(\boldsymbol{x}_{k_j}\right) + \\ \mu_l\left(\boldsymbol{x}_{k_j}, J_{C_0}\left(\boldsymbol{x}_{k_j}\right), V_l\right)\end{matrix}\right)\right\}. \quad (26)$$

Then, based on the set inclusion Definition 3 ($\Psi \subset \mathrm{B} \subset \Xi$ or $\Xi \subset \mathrm{B} \subset \Psi$) and on Lemma 1, if $X_l^+\left(\boldsymbol{x}_{k_j}\right) \cap \mathrm{B} \neq \emptyset$ we have $X_l^+\left(\boldsymbol{x}_{k_j}\right) \cap \Xi = \emptyset$. Then, according to Algorithm 2, $\mathrm{T}_l = \mathrm{T}_{l-1} \cup \{\boldsymbol{x}_{k_j}\}$, such that for all trials $l+1, \ldots, L_{max}$ we get

$$X_{[l+1:L_{max}]}^+\left(\boldsymbol{x}_{k_j}\right) = \left\{\boldsymbol{x}_{k_j+1} \,\middle|\, \boldsymbol{x}_{k_j+1} = F\left(\boldsymbol{x}_{k_j}, C_0\left(\boldsymbol{x}_{k_j}\right) + \mathrm{M}_{[l+1:L_{max}]}\left(\boldsymbol{x}_{k_j}\right)\right)\right\} = X_l^+\left(\boldsymbol{x}_{k_j}\right) \quad (27)$$

This proves that the reachable set for state $\boldsymbol{x}_{k_j}$ is maintained for all subsequent iterations $l+1, \ldots, L_{max}$, avoiding transitions to the unsafe set Ξ. Based on the φ-bounded increase in exploration controller variance between successive l trials, this property extends to all $\boldsymbol{x}_k \in \Psi$ as $l \to L_{max}$, proving that system (14) remains invariant in $\Psi \cup \mathrm{B}$.

2. Appendix B

B.1 Approximate Q-learning algorithm

Using the database DBM_l collected during Algorithm 2, we update the Q-function using

$$\boldsymbol{w}_Q^{i+1} = arg\,\min_{\boldsymbol{w}} \frac{1}{\Pi} \sum_{k=0}^{\Pi-1} \left(\begin{matrix}\tilde{Q}(\boldsymbol{x}_k, \boldsymbol{u}_k, \boldsymbol{w}) - r(\boldsymbol{x}_k, \boldsymbol{u}_k) - \\ \gamma \tilde{Q}\left(\boldsymbol{x}_{k+1}, \tilde{C}\left(\boldsymbol{x}_{k+1}, \boldsymbol{w}_C^i\right), \boldsymbol{w}_Q^i\right)\end{matrix}\right)^2. \quad (28)$$

The controller NN weights at iteration i are updated using

$$\boldsymbol{w}_C^{i+1} = \arg\min_{\boldsymbol{w}} \frac{1}{\Pi} \sum_{k=0}^{\Pi-1} \tilde{Q}\left(\boldsymbol{x}_k, \tilde{C}(\boldsymbol{x}_k, \boldsymbol{w}), \boldsymbol{w}_Q^{i+1}\right) \quad (29)$$

The Approximate Q-learning (AQL) algorithm is described by the following steps.

Algorithm 3. Approximate Q-learning

Input: Collected dataset DBM_l for trial l.

Step 1. Initialize $l_{w_C}, I, \Delta_Q, \iota = 0, i = 0$.

Step 2. Update the Q-function using (28).

Step 3. Initialize $\iota = 0$ and $\boldsymbol{w}^{0,i} = \boldsymbol{w}^i$. Issue I gradient steps using (29). Then, set $\boldsymbol{w}^{i+1} = \boldsymbol{w}^{I,i}$.

Step 4. If $\left\|\boldsymbol{w}_Q^{i+1} - \boldsymbol{w}_Q^i\right\| < \Delta_Q$, stop the algorithm run and use controller $\tilde{C}\left(\boldsymbol{x}_k, \boldsymbol{w}_C^i\right)$. Else, go to Step 2.

B.2 ICVF NN approximation

The ICVF NN approximation learning requires a distinct dataset collected using only $C_0(\boldsymbol{x}_k)$ in closed loop with (1). Let the collected database be $DBM_{C_0} = \{(\boldsymbol{x}_k, \boldsymbol{u}_k, \boldsymbol{x}_{k+1})\}, k = 1{:}\Pi_{C_0}$, where Π_{C_0} is the number of samples. We issue an iterative process with index $\iota = 1, 2, 3, \ldots, I$ and solve

$$\boldsymbol{w}_{J_{C_0}}^{\iota+1} = arg\,\min_{\boldsymbol{w}} \frac{1}{\Pi_{C_0}} \sum_{k=0}^{\Pi_{C_0}} \left(\tilde{J}_{C_0,\iota}(\boldsymbol{x}_k, \boldsymbol{w}) - r(\boldsymbol{x}_k, \boldsymbol{u}_k) - \gamma \tilde{J}_{C_0,\iota}\left(\boldsymbol{x}_{k+1}, \boldsymbol{w}_{J_{C_0}}^\iota\right)\right)^2 \quad (30)$$

The Approximate initial controller Value function algorithm is described by the following steps.

Algorithm 4. Approximate initial controller Value Function

Step 1. Initialize I, Δ_Q, Π_{C_0}, $\iota = 0$. Collect database DBM_{C_0} using the controller $C_0(\boldsymbol{x}_k)$ in closed loop with system (1).

Step 2. Update the Q-function using (30).

Step 3. If $\|\boldsymbol{w}_Q^{\iota+1} - \boldsymbol{w}_Q^{\iota}\| < \Delta_{J_{C_0}}$, stop the algorithm run and use $\tilde{J}_{C_0,\iota}\left(\boldsymbol{x}_k, \boldsymbol{w}_{J_{C_0}}^{\iota}\right)$. Else, go to Step 2.

The convergence of this algorithm using NN approximators was devised in Lala (2025).

3. Appendix C

In this section we analyze how the selection of ρ impacts the detection of B and T_l, and consequently the system safety. The minimal radius ensuring reliable detection is characterized by the next theorem.

Theorem 3. Let Λ be a finite set of points and $\rho_{safe} = max\{\rho > 0 | \forall \boldsymbol{x} \in \Lambda, \exists \boldsymbol{y} \in \Lambda: \|\boldsymbol{x} - \boldsymbol{y}\| < \rho\}$. Let the open ball $BL(\boldsymbol{x}, \rho) = \{\boldsymbol{y} | \|\boldsymbol{x} - \boldsymbol{y}\| < \rho\}$. Then, for any $\rho \geq \rho_{safe}$ and query point $\boldsymbol{x}_k$, $\boldsymbol{x}_k \in \Lambda^{\rho} = \cup_{\boldsymbol{x} \in \Lambda} BL(\boldsymbol{x}, \rho)$ if and only if $KNearNS(\boldsymbol{x}_k, \Lambda, K, \rho_{safe})$ is non-empty.

Proof. By the definition of ρ_{safe}, for any $\boldsymbol{x} \in \Lambda$ there exists $\boldsymbol{y} \in \Lambda$ such that $\|\boldsymbol{x} - \boldsymbol{y}\| < \rho_{safe} \leq \rho$, with $BL(\boldsymbol{x}, \rho) \cap BL(\boldsymbol{y}, \rho) \neq \emptyset$. This creates a chain of overlapping ρ-balls that connects any two points in Λ, making Λ^{ρ} connected. Now, for any $\boldsymbol{x}_k \in \Lambda^{\rho}$, there exists $\boldsymbol{y} \in T_l$ such that $\|\boldsymbol{x}_k - \boldsymbol{y}\| < \rho$, making $KNearNS(\boldsymbol{x}_k, \Lambda, K, \rho)$ non-empty. Conversely, assume $KNearNS(\boldsymbol{x}_k, \Lambda, K, \rho)$ is non-empty, then there exists, $\boldsymbol{y} \in \Lambda$ such that $\|\boldsymbol{x}_k - \boldsymbol{y}\| < \rho$, implying that $\boldsymbol{x}_k \in BL(\boldsymbol{y}, \rho) \subset \Lambda^{\rho}$. ■

The practical selection of ρ_{safe} for both finite sets $\Lambda = T_l$ and $\Lambda = \mathrm{B}$ is made by starting with a small ρ and increasing it until each point in Λ has a neighbor and Λ^{ρ} becomes connected through overlapping balls. For increased conservativeness, a higher value than the found ρ_{safe} can be selected, with the trade-off of limited exploration.

4. Appendix D

D.1 NNs training details

ICVF is approximated using a NN with structure $10 - 100 - 1$. The activation functions are hyperbolic tangent in the hidden layer and linear function at the output layer. Its weights are updated using stochastic gradient run in full-batch mode.

The Q-function and controller are approximated by NNs with architectures of $10 - 50 - 1$ and $8 - 30 - 1$ neurons in each layer. The activation functions are hyperbolic tangent in the hidden layer and linear function at the output layer for both Q-functions and controller NNs. Their weights are updated using stochastic gradient run in full-batch mode.